

Catalan-learning children productively generalize saltatory alternations

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Abstract

Catalan presents an unexpected phonological pattern: while voiced obstruents surface as voiceless word-finally (e.g., ['griz-ə] ~ ['gris] 'grey-F.SG/M.SG'), stems with final /ʒ/ surface as [t̪], skipping over the phonetically intermediate and phonotactically licit [ʃ] (e.g., ['bɔʒ-ə] ~ ['bɔt̪] 'crazy-F.SG/M.SG', *['bɔʃ]). This type of alternation, in which a segment maps to a surface form more phonetically distant than an intermediate but licit candidate, is referred to as saltation. Saltation is formally underivable in classical Optimality Theory and has been shown to be liable to repair and difficult to learn for both adults and children. Whether speakers acquiring a language that actually exhibits saltation can learn it productively remains unknown. In an experiment using the Central Fixation Procedure, Catalan-learning two-year-olds related novel feminine forms to masculine forms with saltation applied, but not to forms with straightforward Final Devoicing applied. Children therefore extend saltation to novel stems and do not repair it, even though Catalan-speaking adults extend it only about half the time. We argue that this trajectory reflects *lexical dilution*. Saltatory stems are few, frequent and acquired early, so their share of the lexicon peaks early and falls as vocabulary grows.

Keywords: Catalan, saltation, productivity, learnability

Word count: 9609

1. Introduction

A central question in phonological theory and language acquisition is the extent to which learners are constrained in the phonological patterns they can acquire. One influential line of research proposes that learners are biased toward learning certain phonological patterns (e.g., Baković, 2000; Wilson, 2003; Moreton, 2008). In many such proposals, Universal Grammar forms the basis for innate biases (see Chomsky & Halle, 1968; Archangeli & Pulleyblank, 1994; Clements & Hume, 1995; Steriade, 2001; Hayes & Steriade, 2004 among many others) which render particular phonological patterns unlearnable or so difficult to learn that they are acquired extremely slowly, imperfectly, or dealt with via memorization.

A particularly strong test case for this hypothesis is saltation (Hayes & White, 2015), a type of phonological alternation in which an underlying segment maps to a surface form that is phonetically more distant than an intermediate, but phonotactically licit, candidate. Experimentally, both adults and infants have been shown to resist saltatory alternations in artificial languages taught in the lab and instead repair them (White, 2014; White & Sundara, 2014). However, saltation does occur in natural languages, raising the question of whether it can be acquired under naturalistic conditions, with long-term exposure. Catalan provides an ideal testing ground for this issue as it exhibits a saltatory alternation whereby stem-final /z/ surfaces as [tʃ] rather than the closer and phonotactically licit [ʃ] in word-final position. Recent wug-testing of adult Catalan speakers has shown that this alternation is productively extended to novel stems, but only in a bit over half of responses despite the pattern being exceptionless in the lexicon (Liang et al., 2024). Whether child learners productively generalize saltatory patterns, and whether the bias against saltation documented in artificial grammar learning (AGL) studies persists in the face of naturalistic input, remains unknown.

The present study addresses these questions by testing Catalan-learning children on their generalization of the saltatory alternation to novel forms. As we will show, they generalize it robustly. This result is initially surprising given the fact that adults only variably extend the alternation. However, as we discuss in our proposal in §3.1, a developmental trajectory of saltation being most productive early in development and declining thereafter can actually be explained by the lexical statistics of the alternation.

1.1 The /z/ ~ [tʃ] alternation in Catalan

The descriptions and transcriptions throughout this paper reflect Central Catalan, the variety spoken in Barcelona and the eastern half of Catalonia, since this is the variety all of the children in this study were acquiring. For expository convenience, rule-based *SPE*-style analyses (Chomsky & Halle, 1968) will be presented below.

Like many languages, Catalan devoices obstruents word-finally, as shown in (1). Note that feminine forms take the suffix [-ə] whereas masculine forms take no ending; throughout this paper, feminine and masculine inflected pairs are used to make the underlying segment of stem-final consonants transparent. The feminine makes the underlying representation of the stems clear as the stems are not word-final.

(1) a. Final Devoicing

[-sonorant] → [-voice] / ___]_{word} ‘Devoice obstruents word-finally’

b. Derivations

‘grey.M.SG’	‘grey-F.SG’	‘fat.M.SG’	‘fat-F.SG’	
/griz/	/griz-ə/	/gras/	/gras-ə/	underlying representations
gris	—	—	—	Final Devoicing
[ˈgris]	[ˈgrizə]	[ˈgras]	[ˈgrasə]	surface representations

Given Final Devoicing, one might expect all voiced obstruents to surface as their voiceless counterparts word-finally. However, stems with underlying final /ʒ/ behave unexpectedly. These stems surface with final [tʃ̥] as in [ˈbɔʒ-ə] ~ [ˈbɔtʃ̥] ‘crazy-F.SG/M.SG’ (not *[ˈbɔʃ]). This is instead of the expected [ʃ] - the voiceless counterpart of /ʒ/, despite it being phonotactically well-formed word-finally (e.g., [ˈfluʃ-ə] ~ [ˈfluʃ] ‘loose-F.SG/M.SG’). The segment /ʒ/ thus “skips over” (*saltates*) the phonetically intermediate and phonotactically legal [ʃ], changing both the voicing and continuancy of the underlying segment simultaneously. One option to account for this alternation in rule-based phonology is to add a /ʒ/-to-[tʃ̥] rule as shown in (2). This rule results in an opaque interaction with Final Devoicing as the Final Devoicing rule counterbleeds /ʒ/-to-[tʃ̥] rule.

(2) a. /ʒ/-to-[tʃ̥] rule

/ʒ/ → [tʃ̥] / ___]_{word} ‘Map /ʒ/ to [tʃ̥] word-finally’

b. Derivations

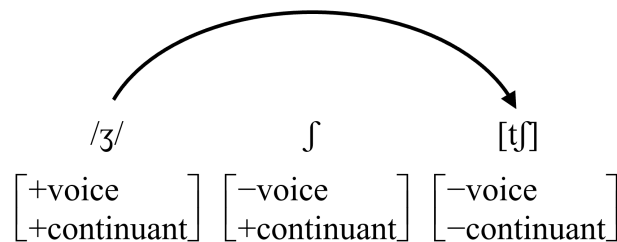
‘crazy.F.SG’	‘crazy.M.SG’	‘grey.M.SG’	‘loose.M.SG’	
/bɔʒ-ə/	/bɔʒ/	/griz/	/fluʃ/	underlying representations
—	bɔtʃ̥	—	—	/ʒ/-to-[tʃ̥] rule
—	—	gris	—	Final Devoicing

[ˈbɔʒə] [ˈbɔtʃ] [ˈɡris] [ˈfluʃ] surface representations

It is worth commenting on why the rule in (2) has to be stated for /ʒ/ specifically. A more phonetically coherent rule would be to affricate word-final palatal fricatives, e.g., /ʃ, ʒ/ → [–continuant] / __]_{word}. Such a rule would be simple and refer to a natural class, but it makes the wrong prediction as it would derive *[ˈflutʃ] from /fluʃ/, whereas the attested masculine form is [ˈfluʃ]. Word-final [ʃ] is robustly attested across many stems in Catalan and surfaces faithfully. The process must therefore pick out the voiced member of the pair and leave the voiceless member alone.

This type of alternation, in which an underlying segment maps to a surface form that is phonetically more distant than an intermediate, but phonotactically licit, candidate, has been called saltation (Hayes & White, 2015), and is illustrated in (2). Note that the term “saltation” describes the pattern in the data rather than a mode of analysis. In this paper, we use saltation as a descriptive label and not as a commitment to a particular theoretical approach.

(3) *Saltation in Catalan*



Saltation has been argued to be “unnatural” as it is never innovated directly, but arises only through a series of historical accidents (Hayes & White, 2015). The Catalan case is consistent with this view. The modern saltatory alternation arose through a chain of sound changes beginning with the palatal stop [tʃ] derived from Vulgar Latin /j/, /dj/, /ge, gi, gj/, /bj/, which later shifted to [dʒ] through increased prominence in the plosive burst (Recasens i Vives, 2023, p. 72). Loss of the final vowel in the masculine, around the middle of the 13th century, left the affricate in word-final position, where it devoiced, yielding [tʃ̥]. Intervocalic deaffrication subsequently produced /ʒ/ in the feminine, yielding the modern surface pairs (e.g., MEDIU → mig [tʃ̥]). The stages are illustrated in (4) with the fictional Vulgar Latin stem /taj/. We note that the relative chronology of intervocalic deaffrication and word-final devoicing is not clearly established. However, we adopt the ordering in (4) for illustrative purposes as devoicing appears to have followed apocope closely. Nevertheless, the saltatory outcome follows from

either ordering since deaffrication is restricted to intervocalic position, so it never affects the word-final affricate.

(4) *Historical origin of saltation in Eastern Catalan, Rossellonese, and Western Catalan for Latin /j, dj, ge, gi, gj, bj, vj/*

Stage	Masculine	Feminine	Change
I	ta _j -u	ta _j -a	Vulgar Latin /j, dj, ge, gi, gj, bj, vj/ > [j]
II	ta _{d̪ʒ} -u	ta _{d̪ʒ} -a	[j] > [d̪ʒ] non-word-finally (word-initial, intervocalic, postconsonantal)
IIIa	ta _{d̪ʒ}	ta _{d̪ʒ} -ə	Loss of the final vowel in the masculine
IIIb	ta _{t̪ʃ}	ta _{d̪ʒ} -ə	Word-final devoicing
IV	ta _{t̪ʃ}	ta _ʒ -ə	Intervocalic deaffrication

The final step in this history is less well understood. At some point after intervocalic deaffrication, intervocalic [d̪ʒ] was reintroduced into the language in some lexical items, as in /'plad̪ʒə/ *platja* 'beach', creating a contrast with non-affricated forms (e.g., /'bɔʒə/ *boja* 'mad, f.') . Exactly how and why this happened is not clear, and several accounts have been proposed. Coromines (1974) proposed that reaffrication was driven phonetically through reinforcement immediately following a stressed vowel, while non-phonetic factors (such as analogy or proclisis) accounted for exceptions before stress (e.g., *pitjor* 'worse') or after stress (e.g., *vagi* 'go.3SG.SBJV'). In contrast, Recasens (2023, p. 74; 2026, pp. 368–377) argues that phonetic factors, while potentially playing a minor role posttonically or adjacent to high vowels, cannot be the primary driver, since intervocalic position cross-linguistically favors lenition rather than fortition. Instead, Recasens treats the reintroduction of [d̪ʒ] as driven primarily by analogy regardless of stress position, either with forms ending in a word-final affricate (e.g., /mat̪ʃ/ *maig* 'May') or with posttonic affricates derived from /tk/ and /dg/ (e.g., /fur'mad̪ʒə/ *formatge* 'cheese'). Regardless of how the intervocalic affricates were reintroduced, the consequence for the learner is the same. Because [d̪ʒ] and [ʒ] both occur between vowels in the modern language, speakers can no longer analyze forms like [bɔʒə] as underlyingly /bɔd̪ʒ-ə/ and must instead analyze them as /bɔʒ-ə/. This yields the modern saltatory alternation.

We note that there is an alternative analysis of this alternation that does not propose underlying /ʒ/ for such forms. The implications of such an analysis will be discussed in §4.3.

1.2 The challenge of saltation

Saltation has been of interest in the literature because it has been difficult to account in terms of phonological theory, and because it is difficult for learners. We take these up in turn, and then draw the prediction they jointly make.

Łubowicz (2002) and Ito & Mester (2003) show that saltation is underivable under classical Optimality Theory (Prince & Smolensky, 1993 with the Faithfulness constraints of McCarthy & Prince, 1995). We present their argument applied to the Catalan case below. To derive $[\text{'b}ɔ\text{t}\widehat{\text{f}}]$ from $/bɔ\text{ʒ}/$, a markedness constraint against word-final voiced obstruents ($*[-\text{son}, +\text{voi}]\#$) must outrank IDENT[voice]. At the same time, a markedness constraint against word-final palatal fricatives (e.g., FORT(pal)# as in Bonet & Lloret (2018), violated once by every word-final $[\text{ʃ}]$ or $[\text{ʒ}]$) must outrank IDENT[cont]. That ranking selects the correct winner for $/bɔ\text{ʒ}/$, as in (5).

(5) *Deriving $[\text{'b}ɔ\text{t}\widehat{\text{f}}]$ from $/bɔ\text{ʒ}/$*

$/bɔ\text{ʒ}/$	$*[-\text{son}, +\text{voi}]\#$	FORT(pal)#	IDENT[voice]	IDENT[cont]
☞ a. $bɔ\text{t}\widehat{\text{f}}$			*	*
b. $bɔ\text{ʃ}$		*!	*	
c. $bɔ\text{ʒ}$	*!	*		

But the same ranking, applied to a stem that is underlyingly $/ʃ/-$ final, derives the wrong output, as in (6).

(6) *Ranking applied to $/flu\text{ʃ}/$*

$/flu\text{ʃ}/$	$*[-\text{son}, +\text{voi}]\#$	FORT(pal)#	IDENT[voice]	IDENT[cont]
☞ a. $flu\text{t}\widehat{\text{f}}$				*
× b. $flu\text{ʃ}$		*!		

In general, whatever markedness constraint forces a change from $/ʒ/$ to $[\text{ʃ}]$ should also prefer $[\text{ʃ}]$ over $[\text{t}\widehat{\text{f}}]$, since the latter incurs an additional Faithfulness violation (continuancy, in addition to voicing).

However, it is possible to analyze saltation by adding additional machinery to the theory. Such approaches work by encoding the pattern’s markedness directly. For example, one approach is to add constraints which penalize specific input-output mappings in the form of *MAP constraints (Zuraw, 2007; Zuraw, 2013; Hayes & White, 2015). A constraint *MAP(x,y) penalizes an input x that surfaces as y. Saltation is then derivable, as in (7) through the ranking *MAP(ʒ,ʃ) ≫ *MAP(ʒ,tʃ). Because *MAP constraints are indexed to particular input–output pairs, neither *MAP(ʒ,ʃ) nor *MAP(ʒ,tʃ) are violated for an input like /fluʃ/

(7) *Deriving saltation with *MAP constraints*

/bɔʒ/	*[–son, + voi] #	*MAP(ʒ,ʃ)	*MAP(ʒ,tʃ)	IDENT[voice]	IDENT[cont]
☞ a. bɔtʃ			*	*	*
b. bɔʃ		*!		*	
c. bɔʒ	*!				

/fluʃ/	*[–son, + voi] #	*MAP(ʒ,ʃ)	*MAP(ʒ,tʃ)	IDENT[voice]	IDENT[cont]
d. flutʃ					*!
☞ e. fluʃ					

It is worth noting that the ranking in (7) states that it is more costly to make a smaller phonetic change than a larger one. Such an ordering is the reverse of what the P-map predicts. Under the P-map (Steriade, 2001, 2009), *MAP constraints receive a default ranking that tracks perceptual distance, so that mappings involving more salient changes are penalized more heavily (Zuraw, 2007). Since [ʒ] is perceptually closer to [ʃ] than to [tʃ], the default is *MAP(ʒ,tʃ) ≫ *MAP(ʒ,ʃ), which predicts the repair /ʒ/ ~ [ʃ] rather than the attested alternation. However, this default is simply a bias the learner brings to the learning process. When the learner’s input provides sufficient evidence, the learner may overcome this initial bias. The theory therefore predicts that saltation should be learnable but difficult, and it leaves open the question of how much evidence is enough.

There are also other analytic options that are possible. Local constraint conjunction (Łubowicz, 1998; 2002) and comparative markedness (McCarthy, 2003) can successfully generate saltation but massively overgenerate typologically unattested grammars (see Hayes & White, 2015 for discussion). Stratal approaches (Bermúdez-Otero, 2019) can also derive saltation with /ʒ/ → [dʒ] applying at one stratum and Final Devoicing at another. Such an

analysis is coherent, but it relocates rather than removes the learning problem. The learner must acquire a stem-level process for which there is no surface evidence anywhere in the language, since $[\widehat{d}\mathfrak{z}]$ never appears as the output of the process at that stratum. We take no position here on which of these analyses is correct, but the important point for our purposes is that saltation is predicted to be difficult to learn regardless of the account.

These predictions are indeed borne out experimentally. Using an AGL paradigm, White (2014) trained adult participants on either a saltatory alternation ($[p] \sim [v]$, $[t] \sim [\delta]$) or a non-saltatory control ($[b] \sim [v]$, $[d] \sim [\delta]$). When subsequently tested on untrained phonetically intermediate segments, participants in the saltatory condition preferentially generalized the alternation to those intermediates - effectively repairing the saltation - at significantly higher rates than controls. Thus, participants who were trained on the $[p] \sim [v]$ were likely to also alternate the intermediate segments $[f]$ and $[b]$ with $[v]$. Crucially, this bias persisted even when participants were given explicit evidence that the intermediate segments did not alternate (White, 2014, Experiment 2), suggesting that the bias against saltation not only affects the generalization to untrained segments but also affects how speakers learn alternations explicitly given in the input.

This anti-saltation bias is also present early in development. White & Sundara (2014) found that 12-month-olds trained on a saltatory alternation similarly repaired the pattern by alternating intermediate segments. Converging evidence comes from work on alternations that are not saltatory but differ in the size of the change involved. Sundara et al. (2021) show that infants acquiring English learn the $/d/ \sim [r]$ alternation before the $/t/ \sim [r]$ alternation, the former involving the smaller change from the stem. Following the P-map principle, learners appear to be biased towards alternations that involve minimal change even with long-term naturalistic exposure to the alternations in question. Together, these findings suggest that the bias against saltation is not a product of extensive linguistic experience but may reflect an innate bias.

AGL paradigms, however, expose participants to novel alternations over short timescales in the lab, and therefore cannot speak to whether sustained, naturalistic exposure to saltation in the input is sufficient to overcome this bias. More direct evidence on this question comes from wug-testing of native Catalan speakers. Liang et al. (2024) wug-tested native speakers of Central Catalan on four stem-final alternations, of which the saltatory $/\mathfrak{z}/ \sim [\widehat{t}\mathfrak{j}]$ alternation was one. The saltation condition comprised ten novel polysyllabic stems, each presented in the feminine (e.g., $[\mathfrak{s}\mathfrak{a}'\mathfrak{l}\mathfrak{o}\mathfrak{z}\text{-}\mathfrak{a}]$, $[\mathfrak{A}\mathfrak{u}'\mathfrak{d}\mathfrak{a}\mathfrak{z}\text{-}\mathfrak{a}]$) with the participants hearing each novel item in isolation and then in a sentence before being asked to produce the corresponding masculine form. In total, their experiment consisted of 37 native speakers, with each participant contributing two saltation trials.

Ultimately, they found that adult Catalan speakers do productively extend the saltatory /ʒ/ ~ [tʃ] alternation to novel stems, though at a rate of 57%, which contrasts with the categorical nature of the phenomenon in the lexicon. This underapplication relative to the lexicon is itself behaviourally consistent with the view that saltation is marked - even after a lifetime of exposure to it in the input. Crucially, when speakers in the study did repair saltation, the predominant strategy was /ʒ/ ~ [ʃ] (underapplying the /ʒ/-to-[tʃ] rule) despite the fact that forms with a /ʒ/ ~ [ʃ] alternation are entirely unattested in the Catalan lexicon.

Whether children acquiring Catalan (or any other language with saltation) can similarly acquire the alternation productively or whether they undergo a stage where they repair the alternation, however, remains unknown.

1.3 Saltation in Catalan child-directed speech

Presumably, a necessary prerequisite for acquiring saltation naturalistically is sufficient exposure to the alternation in the input. Although saltation occurs in relatively few stems in the Catalan lexicon (approximately 15, based on Wheeler, 2005; Liang et al., 2024), several of these are among the most frequent words in the language, meaning that children are likely to encounter the alternation repeatedly and early. To assess this, we examined all Catalan corpora available in CHILDES (MacWhinney, 2000), comprising six corpora of child-directed speech totalling 247,709 words (Bel, 1998; Llinàs-Grau, 2000; Serra et al., 2000; Llinàs-Grau & Coll-Alfonso, 2001; Aparici, 2010; Esteve-Gibert, 2010).

Across the corpora, we identified ten stems in which /ʒ/ alternates with [tʃ]. These included various forms of *fer* 'to do', *haver* 'to have', *passejar* 'to walk', *veure* 'to see', and *anar* 'to go'. The dominance of a small number of high-frequency verbs is striking: *vaig* [batʃ] 'I go', *veig* [betʃ] 'I see', and *faig* fa[tʃ] 'I do' alone account for 720 of the 769 tokens of [tʃ]-final forms. In total, [tʃ]-final saltatory forms constitute 0.3% of the entire tokens in the corpus, alongside 103 tokens of corresponding [ʒ]-final forms from the same alternating paradigms.

Table 1. *Frequency of saltation in Catalan child-directed-speech*

[tʃ] form	[tʃ] Tokens	[ʒ] form(s)	[ʒ] Tokens	Translation
bate[tʃ]	1	bate[ʒ]at	1	'baptism'
bo[tʃ]	4	bo[ʒ]a	1	'crazy'
fa[tʃ]	104	fa[ʒ]is	2	'do'
fu[tʃ]	5	fu[ʒ]ir	1	'run away'

ha[t̪]	22	ha[ɜ]i, ha[ɜ]is, ha[ɜ]in	14	‘have ‘ (aux.)
mare[t̪]	1	mare[ɜ]ada, mare[ɜ]arà, mare[ɜ]aràs, mare[ɜ]o	9	‘to be dizzy’
passe[t̪]	14	passe[ɜ]em, passe[ɜ]es, passe[ɜ]a, passe[ɜ]ant, passe[ɜ]ar	20	‘walk’
ra[t̪]	2	ra[ɜ]a	1	‘ray, flow’
va[t̪]	484	va[ɜ]i, va[ɜ]is, va[ɜ]in, va[ɜ]a	48	‘go’
ve[t̪]	132	ve[ɜ]i, ve[ɜ]is	6	‘see’
Total	769		103	

While saltatory alternations are thus infrequent in absolute terms, their concentration in high-frequency verbs means that children are likely to encounter them repeatedly and early. These figures suggest that saltation represents a more substantial portion of the early lexicons of Catalan-acquiring children than its low type frequency in the adult lexicon might imply, making it feasible that children may already be sensitive to the alternation.

These figures make clear that Catalan-learning children receive consistent, early exposure to saltatory alternations in their input, providing a basis for acquisition of the pattern. The question of whether this exposure is sufficient is what we seek to answer directly in the experiments that follow.

1.4 Experimental overview

In the present study, we report an experiment using the Central Fixation Procedure (Cohen, 1972), testing two groups of Catalan-learning children with a mean age of approximately 24 months. Children in both groups were familiarized with novel feminine forms ending in [ɜ-ə] ([kɛɜə], [ʌɜə], [təɜə], [ruɜə]) and then tested on the corresponding masculine forms. The groups differed only in the masculine forms presented at test. In the Saltation condition, these forms had saltation applied ([kɛt̪], [ʌt̪], [tat̪], [rut̪]); in the Final Devoicing condition, they had straightforward Final Devoicing applied ([kɛʃ], [ʌʃ], [taʃ], [ruʃ]). Condition was manipulated between subjects.

The Saltation condition tests whether Catalan-learning children productively generalize the /ɜ/ ~ [t̪] alternation to novel words. If children have acquired saltation as part of their phonological grammar, they should be able to relate novel feminine forms ending in [ɜ-ə] to

their corresponding masculine forms ending in $[\widehat{tʃ}]$. By contrast, if the bias against saltation documented in AGL studies (White, 2014; White & Sundara, 2014) is strong enough to overcome the saltatory input children have received, they should fail to relate such forms. Crucially, a finding of successful generalization would demonstrate not only that saltation is learnable by children, but also that naturalistic exposure to saltation in the input is sufficient to override the bias against saltation. Such a result would suggest that the saltatory alternation is represented productively as part of the children’s phonological grammar rather than merely as a set of memorized lexical pairs.

However, a familiarity preference in the Saltation condition on its own would be compatible with two alternative explanations, and the Final Devoicing condition is designed to rule both of them out. Firstly, children might be relating the familiar feminine forms to the masculine forms not on the basis of the $/ʒ/ \sim [\widehat{tʃ}]$ alternation specifically, but simply on the basis of shared phonological material. For instance, they may relate $[kɛʒə]$ to $[kɛ\widehat{tʃ}]$ because both begin with $[kɛ]$, rather than because they have learned the relevant alternation. Although prior work has shown that infants do not arbitrarily relate forms with comparable phonological overlap (e.g., Kim & Sundara, 2021; Liang & Sundara, 2025), we include the Final Devoicing condition as a direct test of this possibility given that previous work has primarily been done on English-learning infants.

Secondly, even if children indeed learn the $/ʒ/ \sim [\widehat{tʃ}]$ alternation, they may not have acquired saltation. Instead, they may be repairing saltation, as has been reported for infants given short-term laboratory exposure to saltatory alternations (White & Sundara, 2014), by underapplying the $/ʒ/-to-[\widehat{tʃ}]$ rule. If this is the case, we would expect the children to accept $[\ʃ]$ forms as familiar masculine counterparts just as readily as the $[\widehat{tʃ}]$ forms. The particular repair pattern that we test is independently motivated as $/ʒ/ \sim [\ʃ]$ is the outcome predicted by straightforward application of Final Devoicing, it is the repair predicted by the P-map, and it is the predominant repair strategy produced by Catalan-speaking adults (Liang et al., 2024).

2. Experimental Overview

This experiment tested whether Catalan-learning children extend the $/ʒ/ \sim [\widehat{tʃ}]$ alternation to novel forms. Children were familiarized with novel feminine forms ending in $[ʒ-ə]$ and then tested on masculine forms that had either saltation applied (the Saltation condition) or straightforward Final Devoicing applied (the Final Devoicing condition). If they have acquired saltation as part of their phonological grammar, they should relate the familiarized feminine forms to their corresponding masculine forms in the Saltation condition, but not in the Final Devoicing condition. If the anti-saltation bias documented in AGL studies is strong enough to

override two years of naturalistic input, they should fail to relate the forms in the Saltation condition.

2.1 Stimuli

Four novel phonotactically-legal feminine target words were used: [kɛʒə], [ʎəʒə], [taʒə], [ruʒə]. The novel words were constructed to vary both in vowel quality and initial consonant to minimize the potential for item-specific effects. The feminine nonce words were embedded in four six-sentence passages (see Kim & Sundara, 2021; Sundara et al., 2021; Liang & Sundara, 2025 for infant studies using a similar approach), and each nonce word was preceded by the feminine article *la* ‘the’ to assist with segmentation. The passages in Catalan and their English translations are given in Table 2. The masculine test forms differed between the two conditions. In the Saltation condition they had the saltatory alternation applied ([kɛt̪], [ʎət̪], [tat̪], [rut̪]), and in the Final Devoicing condition they had straightforward Final Devoicing applied ([kɛʃ], [ʎəʃ], [taʃ], [ruʃ]). In both conditions the masculine forms appeared in isolation in lists, preceded by the masculine article *el* ‘the’.

In addition, both word-final [t̪] and [ʃ] are well attested word-finally in child-directed speech. Across the same CHILDES corpora described in §1.3, word-final [ʃ] occurs in 27 types and word-final [t̪] in 18, across all [t̪]-final forms rather than only those in saltatory paradigms. The test items in the two conditions are therefore matched on the phonotactic status of their final segment.

The stimuli were recorded by a phonetically trained female native speaker of Central Catalan. She was instructed to read the passages in an animated voice as if she were talking to a young child. The recordings were done in a soundproof booth using an Audio-Technica AT2050 microphone. All stimuli were digitized at a sampling frequency of 22,050 Hz and 16-bit quantization. The average intensity, duration, and pitch of all of the recordings were equalized using Praat (Boersma & Weenink, 2026).

The passages each had a duration of 21.3 s with an average pitch of 211.1 Hz (range 58-598), and their intensity was normalized to 72 dB. The average duration of each of the nonce words in the passages was 625 ms (SD = 88). For the Saltation test lists, the average duration was 748 ms (SD = 73), average pitch was 250 Hz (SD = 64), and average intensity was 74.6 dB (SD = 2.5). For the Final Devoicing test lists, the average duration was 737 ms (SD = 69), average pitch was 259 Hz (SD = 57), and average intensity was 74.9 dB (SD = 1.5). Detailed acoustic analyses can be found on the [Zenodo site](#).

Table 2. *Nonce words and passages used in the experiments*

Nonce word	Passage (Catalan)	Passage (Translation)
<i>keja</i> [keʒə]	La keja petita juga amb un ós. Un ocell canta per a la keja . El gat dorm content amb la keja . La keja bonica riu amb la nena. Els estels brillen damunt la keja . El sol escalfa el jardí de la keja .	The little keja plays with a bear. A bird sings for the keja . The cat sleeps happily with the keja . The pretty keja laughs with the girl. The stars shine above the keja . The sun warms the garden of the keja
<i>lloja</i> [ləʒə]	La lloja verda corre pel jardí. Un conill menja herba amb la lloja . Els nens ballen contents amb la lloja . La lloja valenta juga amb un gos. La lluna surt i saluda la lloja . Un pastís dolç espera la lloja .	The green lloja runs through the garden. A rabbit eats grass with the lloja . The children dance happily with the lloja . The brave lloja plays with a dog. The moon comes out and greets the lloja . A sweet cake awaits the lloja .
<i>taja</i> [taʒə]	La taja vermella canta cada dia. Un drac vola amb la taja . La vaca balla i mira la taja . La taja maca somriu a la nena. Un cargol passeja i saluda la taja . La pluja cau damunt la taja .	The red taja sings everyday. A dragon flies with the taja . The cow dances and looks at the taja . The cute taja smiles with the girl. A snail walks by and greets the taja . The rain falls on the taja .
<i>ruja</i> [ruʒə]	La ruja baixa juga amb pilotes. El tren passa pel costat de la ruja . La ruja bonica somriu al papa. Un peixet fa bombolles per la ruja . Un ocell vola amb la ruja . La ruja dolça juga amb les flors.	The short ruja plays with balls. The train passes beside the ruja . The cute ruja smiles at daddy. A fishy blows bubbles for the ruja . A bird flies with the ruja . The sweet ruja plays with the flowers.

2.2 Participants

The data from 40 Catalan-learning children were included in the analysis, 20 in each condition. In the Saltation condition, the 20 children (12 male, 8 female) had a mean age of 724 days

(range 391–953; SD = 161). In the Final Devoicing condition, the 20 children (12 male, 8 female) had a mean age of 821 days (range 451–1092; SD = 166). Four additional children were tested but were ultimately excluded from the analysis due to fussiness.

All children were required to have at least 50% of their language input as Catalan (assessed via a language questionnaire; Sundara & Scutellaro, 2009). Mean Catalan input was 88% (range 68–100; SD = 14). All children were screened via parental report to ensure no history of speech, language, or hearing difficulties, and were in good health on the day of testing.

The children were recruited through various preschools in the Barcelona area with testing being conducted *in situ* at the child's preschool or home. At least one parent or guardian was present with the child during the testing session. Each family received a USD\$15 electronic gift card as compensation.

2.3 Procedure

Testing was conducted using the Central Fixation Procedure (Cohen, 1972) adapted for the preschool or home environment. It took approximately 10 minutes to test each participant. Testing took place in a quiet dimly-lit room in the preschool or home. During testing, the child sat on a caretaker's lap facing an external monitor in a quiet, dimly-lit room. A webcam placed above the monitor recorded the child's gaze, and stimulus presentation was controlled via Habit (Oakes et al., 2019) on a computer behind the monitor, obscured from the child's view by a large black posterboard. Both the experimenter and caretaker wore noise-cancelling headphones with music playing to prevent any unconscious influence on child looking behaviour.

At the beginning of each trial, a looming circle appeared on the external monitor along with a baby giggle to attract the child's attention to the display. Once the child's attention was fixated on the screen, a Tetris-like video animation was played on the screen along with the auditory stimuli. During the study, the experimenter observed the child through the webcam and recorded the duration of the child's gaze towards the monitor. The presentation of auditory stimuli was completely contingent on the child's looking behaviour, serving as a proxy for listening time - the dependent variable in this study. The trials stopped when the child looked away for more than 2 seconds or at the end of the trial. A minimum look time of 1 second was required for a trial to be counted.

The presentation of the auditory stimuli was split into two phases - a familiarization phase and a test phase. During the familiarization phase, the children were familiarized with two out of four possible novel feminine words (either [kɛʒə], [ʌʒə] or [taʒə], [ruʒə])

embedded in passages until they accumulated 90 seconds of total listening time across the two nonce words. The children were randomly assigned to one of two conditions with half of the children being familiarized on passages containing [kɛʒə] and [ʌʒə] while the other half were familiarized on passages with [taʒə] and [ruʒə]. By splitting the participants into two separate conditions, we sought to further minimize the potential for any item-specific effects.

In the test phase, the children heard lists of all four masculine forms in three blocks for a total of 12 trials. Children in the Saltation condition heard the forms with the saltatory alternation applied ([kɛt̪], [ʌt̪], [tat̪], [rut̪]), and children in the Final Devoicing condition heard the forms with straightforward Final Devoicing applied ([kɛʃ], [ʌʃ], [taʃ], [ruʃ]). Because children were never presented with the masculine forms during familiarization, two of the four masculine forms were potentially familiar to them only if they related those forms to the feminine forms they had heard. The other two masculine forms are completely unfamiliar, as the children never heard their feminine counterparts during the familiarization phase. For example, for children familiarized on [kɛʒə] and [ʌʒə], [kɛt̪] and [ʌt̪] are potentially familiar if they extend saltation whereas [tat̪] and [rut̪] are completely unfamiliar.

2.4 Analysis

Listening times were analyzed in a linear mixed-effects model fitted with the lmerTest package in R (Kuznetsova et al., 2017). Listening time was the dependent variable, with Block (within-subjects; 1-3), Condition (between-subjects; Saltation vs. Final Devoicing), and Trial Type (within-subjects; potentially familiar vs. unfamiliar) and all their interactions as fixed effects.

The model also included a random intercept for Subject to control for differences in the baseline listening times and random slopes for Block by Subject. A model with an additional random slope for Trial Type by Subject was also tested but failed to converge and was not used. The anova() function was used to evaluate fixed effects against the full model. Then, we conducted planned comparisons to examine significant interactions using the emmeans package (Lenth & Piaskowski, 2026).

2.5 Results

The looking time data are presented in Figure 1. The analysis revealed a significant main effect of Trial Type ($F(1, 354) = 15.88, p < 0.001$) and a significant interaction between Condition and Trial Type ($F(1, 354) = 9.91, p = 0.002$), indicating that the effect of Trial Type differed between the two conditions. There was also a significant main effect of Block ($F(2, 38) =$

20.24, $p < 0.001$), reflecting the expected decrease in listening times over the course of the experiment. No other effects reached significance, though we note a marginal three-way interaction between Block, Condition, and Trial Type ($F(2, 354) = 2.98, p = 0.052$). Given that this effect does not reach significance and our key Condition and Trial Type interaction is robust, we do not interpret it further.

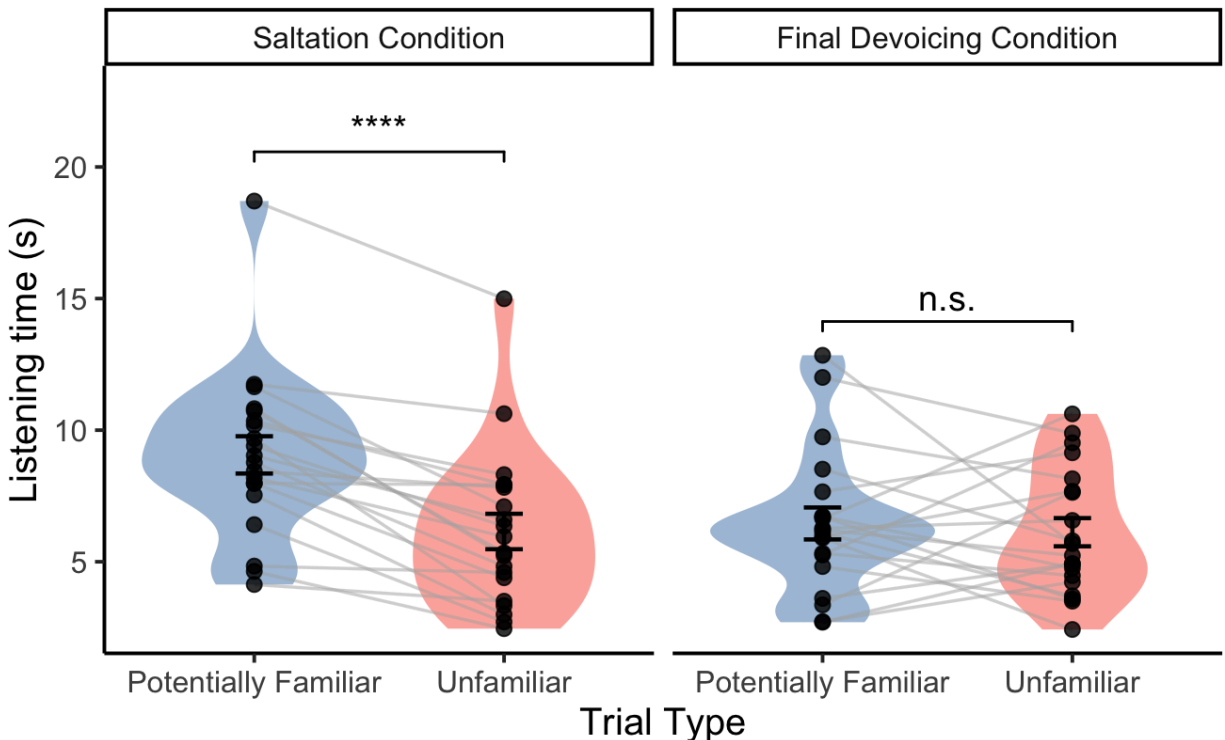


Figure 1. Mean listening times (in seconds) in the Saltation and Final Devoicing conditions by Trial Type. Each dot represents the listening times of an individual participant.

Planned comparisons revealed that, in the Saltation condition, children listened significantly longer to potentially familiar than unfamiliar items ($t(355) = 5.03, p < 0.0001$). This indicates that children successfully related the familiarized feminine forms to their corresponding masculine forms with the saltatory alternation applied, demonstrating productive generalization of the saltatory /ʒ/ ~ [tʃ] alternation to novel stems. By contrast, in the Final Devoicing condition, planned comparisons revealed no significant difference in listening times between potentially familiar and unfamiliar items ($t(352) = 0.59, p = 0.55$), indicating that children did not relate the familiarized feminine forms to masculine forms with Final Devoicing applied, despite word-final [ʃ] being as well attested as word-final [tʃ] in their input (§2.1).

Together, these results demonstrate that Catalan-learning children specifically acquired the saltatory alternation. Had children been matching forms on the basis of shared onset material alone, or repairing saltation by underapplying the /ʒ/-to-[tʃ] rule, a familiarity preference in the Final Devoicing condition would be expected as well. The absence of such a preference rules out both alternatives, providing strong evidence that the Saltation condition results reflect productive acquisition of saltation specifically.

3. General Discussion

This study investigated whether Catalan-learning children can acquire and generalize a saltatory alternation. In the Saltation condition, children generalized the /ʒ/ ~ [tʃ] alternation to entirely novel forms, as shown by their longer listening times to the potentially familiar items. Conversely, in the Final Devoicing condition, the children showed no preference for forms reflecting a potential saltation repair strategy (/ʒ/ ~ [ʃ]). Taken together, these results provide the first evidence that children exposed to a language containing saltation learn the alternation productively, and do not repair it.

The absence of a familiarity preference in the Final Devoicing condition rules out two alternative explanations to these results. First, children are unlikely to be matching forms based solely on shared phonological material since this would predict similar responses to [ʃ]-final forms. Second, children are not repairing saltation by underapplying the /ʒ/-to-[tʃ] rule (e.g., *keja* /kɛʒ-ə/ → *keix* [kɛʃ]). Taken together, these results suggest that Catalan-learning children do indeed generalize the saltatory alternation to novel forms.

These findings bear directly on claims in the literature that saltation is either dispreferred or unlearnable. Prior AGL studies (White, 2014; White & Sundara, 2014) have reported that both adults and infants tend to repair saltatory alternations. However, one important difference between the present study and previous studies is the nature of the input. AGL studies expose learners to novel patterns over short timeframes, whereas the present study has the benefit of learners having been exposed to sustained, naturalistic input demonstrating the saltation pattern. Thus, the results of the present study show that biases against saltation do not prevent acquisition under naturalistic input. Instead, they support a view in which this bias is gradient and can be overcome given enough input to the contrary.

In this respect, our results parallel other cases where learners may favour certain patterns that are deemed more natural but ultimately can acquire “unnatural” patterns given sufficient evidence (e.g., Seidl & Buckley, 2005; Hayes et al., 2009; Baer-Henney & van de Vijver, 2012; Moreton & Pater, 2012a; Moreton & Pater, 2012b; White, 2017; Glewwe, 2019; Siah, 2024; Zheng & Do, 2025). At the same time, our results do not imply that saltation is

unmarked. Adult Catalan speakers show underapplication and occasional repair in production of saltation to novel words, suggesting that the pattern remains somewhat disfavoured even after extensive exposure. Children’s success therefore reflects learnability, but not necessarily equal status with more phonetically natural alternations.

3.1 “Lexical dilution” and the developmental trajectory of saltation

The comparison between the present results and the adult data of Liang et al. (2024) presents an unexpected finding. Two-year-olds extend the alternation to novel stems and show no sign of saltation repair. Adults, with decades of exposure to the same exceptionless pattern, extend it on only 57% of wug trials and produce the unattested [ʃ] on most of the rest. This result runs counter to the usual examples of U-shaped learning in the literature. In the classic case of U-shaped learning, English-learning children initially produce high-frequency irregular past tenses correctly (e.g., *went*, *broke*), then begin overregularizing them (*goed*, *breaked*), before their performance finally recovers to its initial levels (Marcus et al., 1992). In the Catalan case, the learner’s performance instead declines and never manages to recover to its early success.

Some caution is required before treating this as a fact. The two studies use different tasks: a looking-time measure of recognition in the child case and elicited production for adults. As looking-time preferences and production rates are not comparable since looking-time is a much more coarse measure, we cannot say that children generalize saltation more than adults in any quantitative sense. However, what we can say is that, despite the variability in the adult’s willingness to extend saltation to novel forms, young children are readily able to do so. The use of novel forms in our experiment also means that children are able to extend saltation productively and are not just memorizing the set of saltatory forms.

Regardless of the difference in the two studies, we believe that the difference between the adult and child performance is genuine, and we propose that it is a natural consequence of the composition of the learner’s lexicon throughout development. The central tenet behind our proposal is that the ability for a learner to treat an alternation as productive is not determined by the *absolute* amount of evidence that they have encountered, but rather by the *proportion* of examples present in their lexicon.

Three properties of saltation in the Catalan lexicon allow us to tease apart these two possibilities. Firstly, the set of words that undergo saltation in Catalan is small (only around 15 items). Secondly, the words that do undergo saltation are among the highest frequency in the language (e.g., *vaig* ‘I go’, *veig* ‘I see’, *faig* ‘I do’, *haig* ‘I have to’, see Table 1). Thirdly, as a consequence of the second fact, the saltatory forms are among the earliest words acquired by

the learner. Thus, a Catalan-learning two-year-old already knows most of the saltating stems they will ever know.

These facts put together mean that saltation's share of the learner's lexicon peaks fairly early in development and declines monotonically thereafter. A two-year-old with a vocabulary of a few hundred words knows a lexicon in which several saltating stems are present and represent a sizable proportion of the lexicon. An adult with a vocabulary of tens of thousands of words knows the same handful of saltating stems, but these forms are now diluted among thousands of words that either say nothing about the alternation or, in the case of the many non-alternating [ʃ]-final stems, potentially argue against extending it. We dub this phenomenon *lexical dilution*.

The learning process can be viewed as a competition between two different generalizations which are both active in the saltatory forms: the narrow /ʒ/-to-[tʃ] rule, and general Final Devoicing. The narrow rule has perfect *reliability* (the percentage of forms that could undergo a rule which actually do) at every stage of acquisition as no /ʒ/-final stem ever fails to alternate with [tʃ]. However, its *scope* (the number of forms a rule applies to) remains stagnant at around 15 items. On the other hand, Final Devoicing also has perfect reliability as it is never violated in the lexicon, but its scope is every obstruent-final stem in the language which grows dramatically throughout development. In a model which is sensitive to the scope of a particular rule in the learner's current lexicon such as the Minimal Generalization Learner (Albright & Hayes, 2003) or the Tolerance Principle (Yang, 2016), the narrow /ʒ/-to-[tʃ] rule is competitive when it is first supported by a relatively large amount of data compared to Final Devoicing. As the lexicon grows, the general Final Devoicing rule's advantage grows and since the P-map already favours the output of straightforward Final Devoicing, the productivity of the saltatory alternation decreases.

Under this proposal, the infant AGL results of White & Sundara (2014), Catalan child results of this paper, and Catalan adult results of Liang et al. (2024), can all be viewed as different points along the same learning trajectory. In AGL, learners only get minutes of exposure to the pattern, and the anti-saltation bias wins outright. In naturalistic acquisition at two years, the supporting evidence is at its maximum relative proportion, and the bias loses. In adulthood, the same supporting evidence has undergone lexical dilution, and the anti-saltation bias resurfaces.

This account also makes predictions beyond Catalan. Any alternation instantiated by a small set of high-frequency, early-acquired items should show the same profile of maximal productivity in early childhood, declining thereafter. For example, English plural formation offers a promising case. A small set of nouns with stem-final /f/ voice the fricative before the plural suffix (*leaf* ~ *leaves*, *knife* ~ *knives*, *wife* ~ *wives*, *half* ~ *halves*, *wolf* ~ *wolves*). This set

has many similar properties to the Catalan alternation. It is a small set of high-frequency words that are likely to be acquired early by children. The set of non-alternating English plurals grows steadily as the learner's vocabulary expands, and lexical dilution therefore predicts that fricative voicing is at its most productive in early childhood and loses ground thereafter.

The existing research on this topic is unclear. Although Berko (1958) wug-tested children on the nonce form *heaf* and found that they did not voice the fricative in the plural, the children she tested were between four and seven years old and already had substantial vocabularies. Under the lexical dilution account, the window in which the alternating set holds its largest share of the lexicon has passed well before that age range. Thus, testing the prediction of lexical dilution would require an experiment with children at a much younger age.

More generally, testing our proposal means studying the learner's lexicon at each point in development. If productivity depends on the proportion of the current lexicon that supports a generalization, what is needed is an estimate of which words a child of a given age is likely to know and how many of them bear on the alternation in question. Models sensitive to the scope of a generalization, such as the Minimal Generalization Learner or the Tolerance Principle, could then be run over varying sizes of lexicons. This would yield a predicted productivity trajectory for a given alternation, which could be tested against behaviour at each age.

3.2 An alternative repair strategy

In principle, another possible repair strategy would be to alternate the phonetically intermediate segment /ʃ/ with [tʃ̥]. Under such an approach, the mapping would no longer be saltatory, as /ʒ/ would no longer “skip over” /ʃ/ as both /ʒ/ and /ʃ/ would alternate with [tʃ̥]. We did not directly test this possibility for several reasons.

Firstly, such a pattern lacks support in the Catalan lexicon. While the /ʒ/ ~ [tʃ̥] alternation and Final Devoicing process are attested and exceptionless in the lexicon, there is no evidence for a /ʃ/ ~ [tʃ̥] alternation. On the contrary, word-final [ʃ] is robustly attested across many types and surfaces faithfully (e.g., [ˈfluʃ] ‘loose.M.SG’), providing strong evidence that /ʃ/ does not alternate. From a learning perspective, further alternating /ʃ/ with [tʃ̥] would therefore require overriding the well-attested exceptionless pattern in the lexicon.

Secondly, this hypothesized repair is not independently motivated by phonological processes in Catalan. Unlike /ʒ/ ~ [ʃ], which follows directly from Final Devoicing, there is no process that would derive /ʃ/ ~ [tʃ̥]. As such, it is less likely that learners would posit such a mapping.

Thirdly, adult behaviour provides no indication that this repair strategy is available. Although Liang et al. (2024) do not directly test speakers on wug items with [ʃ]-final stems, informal elicitation with native Catalan speakers conducted by the authors yielded no instances of /ʃ/ ~ [tʃ] alternations, with speakers consistently rejecting such forms. While these observations are not based on systematic experimentation, they converge with the lexical and experimental evidence in suggesting that this repair strategy is not available to learners.

Finally, experimental evidence from artificial grammar learning provides additional reason to doubt that this repair is likely. Smolek & Kapatsinski (2018) directly investigate multiple saltation repair strategies by using a free response production task; namely, overgeneralization of the saltatory alternation to the intermediate segment (e.g., repairing saltation by also alternating /ʃ/ ~ [tʃ] in Catalan) and underapplication of the saltatory alternation (e.g., preferring more faithful outputs such as /ʒ/ ~ [ʃ] in Catalan). They ultimately find that the primary response to saltatory alternations was underapplication with reduced productivity of the alternation rather than extension to intermediate segments. This result, whereby saltatory alternations lose productivity rather than spread to intermediate sounds, is also consistent with the fate of the English [k] ~ [s] alternation in pairs like *electri[k]/electri[s]ity*, which skips over intermediate [t] and appears to have been repaired through a gradual loss of productivity rather than overgeneralization to [t] (Pierrehumbert, 2006).

The convergence of lexical, behavioural, and experimental evidence therefore suggests that it is unlikely that the /ʃ/ ~ [tʃ] repair is available to learners, and we leave its direct investigation to future work.

3.3 Alternative analyses of the alternation

As noted in §1.1, we have adopted saltation as a descriptive label for the Catalan alternation without commitment to a particular theoretical analysis. However, there are analyses that propose that the Catalan alternation is non-saltatory. One influential set of analyses proposes that the underlying form of stems of pairs like [bɔʒ-ə] ~ [bɔtʃ] should be /bɔdʒ/, with surface [ʒ] derived by a process of intervocalic spirantization and [tʃ] by straightforward Final Devoicing (Torres-Tamarit, 2016; Bonet & Lloret, 2018).

To account for the fact that Catalan has no [ʒ] ~ [ʃ] alternations, this type of analysis requires additional apparatus. Some mechanism, such as stem structure constraints, must ensure that no stem is underlyingly /ʒ/-final so that the grammar does not produce the unattested [ʒ] ~ [ʃ] alternation. In other words, this analysis requires enforcing the generalization that all stem-final phonetic [ʒ]s are derived from underlying /dʒ/ rather than

/ʒ/. This is precisely the "free ride" problem discussed by McCarthy (2005) where learners must deduce that all surface [ʒ]s derive from /d͡ʒ/ rather than /ʒ/ even in the absence of direct alternation evidence (e.g., the learner only sees surface pairs like [bɔʒə] ~ [bɔt͡ʃ] and has to deduce that the underlying form for the stem should be /bɔd͡ʒ/ despite [d͡ʒ] never occurring in any surface forms). McCarthy (2005) shows that arriving at such a generalization of the form "all [X]s are derived from /Y/" is itself a serious learning problem.

This alternative analysis also casts the adult wug test data in a different light. If the stems of this type are underlyingly /d͡ʒ/-final, then adults who produce [ʃ] for a novel [ʒ]-final stem are not repairing saltation at all. Instead, they are mislearning the underlying representation as /ʒ/ before applying straightforward Final Devoicing, meaning the adults have failed to take the "free ride". The locus of adult variability is, thus, a result of the underlying representation itself rather than the phonological grammar. Nevertheless, this result would still be compatible with the lexical dilution hypothesis. The generalization that [ʒ] derives from /d͡ʒ/ is itself supported by a small handful of frequent, early-acquired forms which is diluted as the learner's lexicon grows, and the children in the present experiment are still performing a feat that adults only did about half of the time.

We therefore remain agnostic about whether the Catalan alternation is best analyzed as saltation or via the /d͡ʒ/ underlying form, and our results do not distinguish between the two alternatives. What our results do demonstrate is that, no matter the account, children have acquired a marked phonological generalization. If the saltatory analysis is correct, they have learned a marked alternation that has been argued to be formally underivable and pose learnability challenges. If the /d͡ʒ/ analysis is correct, they have successfully taken a "free ride" on the /d͡ʒ/ ~ [ʒ] unfaithful map, generalizing it to nonalternating stems without direct alternation evidence - a learning feat that McCarthy (2005) identifies as independently challenging.

4. Conclusion

In sum, this study provides the first evidence that children exposed to a language with a saltatory alternation can actually learn the alternation. Despite prior claims that saltation is unlearnable or subject to repair, Catalan-learning children generalize the pattern robustly and do not repair it by underapplying the /ʒ/-to-[t͡ʃ] rule. These findings challenge models that treat saltation as formally underivable and instead support approaches to phonological learning in which phonological biases are gradient and can be overridden given robust evidence in the input.

Our results also suggest that the amount of evidence required to overcome a particular bias is best measured relative to the learner's current lexicon rather than in absolute terms. On

the account we have proposed, an alternation carried by a small set of very frequent words is at its most productive early, when those words make up a large share of what the child knows, and loses ground through lexical dilution as other forms increase their share in the lexicon.

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Competing interests

The authors have no competing interests to declare.

Author contributions

Kevin Liang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Reviewing & Editing, Visualization, Supervision, Project Administration, Funding Acquisition.
Gemma Repiso Puigdelliura: Investigation, Resources, Writing - Original Draft, Writing - Reviewing & Editing, Project Administration.

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